

Economics 2220

Principles Of Macroeconomics

Professor Zhigang Feng

Lecture 7:

Economic Growth

► What you will learn in this chapter



- Why long-run economic growth is measured as the increase in real GDP per capita, how real GDP per capita has changed over time, and how it varies across countries
- Why productivity is the key to long-run economic growth and how productivity is driven by physical capital, human capital, and technological progress
- The factors that explain why long run growth rates differ so much among countries
- How growth has varied among several important regions of the world and why the convergence hypothesis applies to economically advanced countries
- The question of sustainability and the challenges to growth posed by scarcity of natural resources and environmental degradation

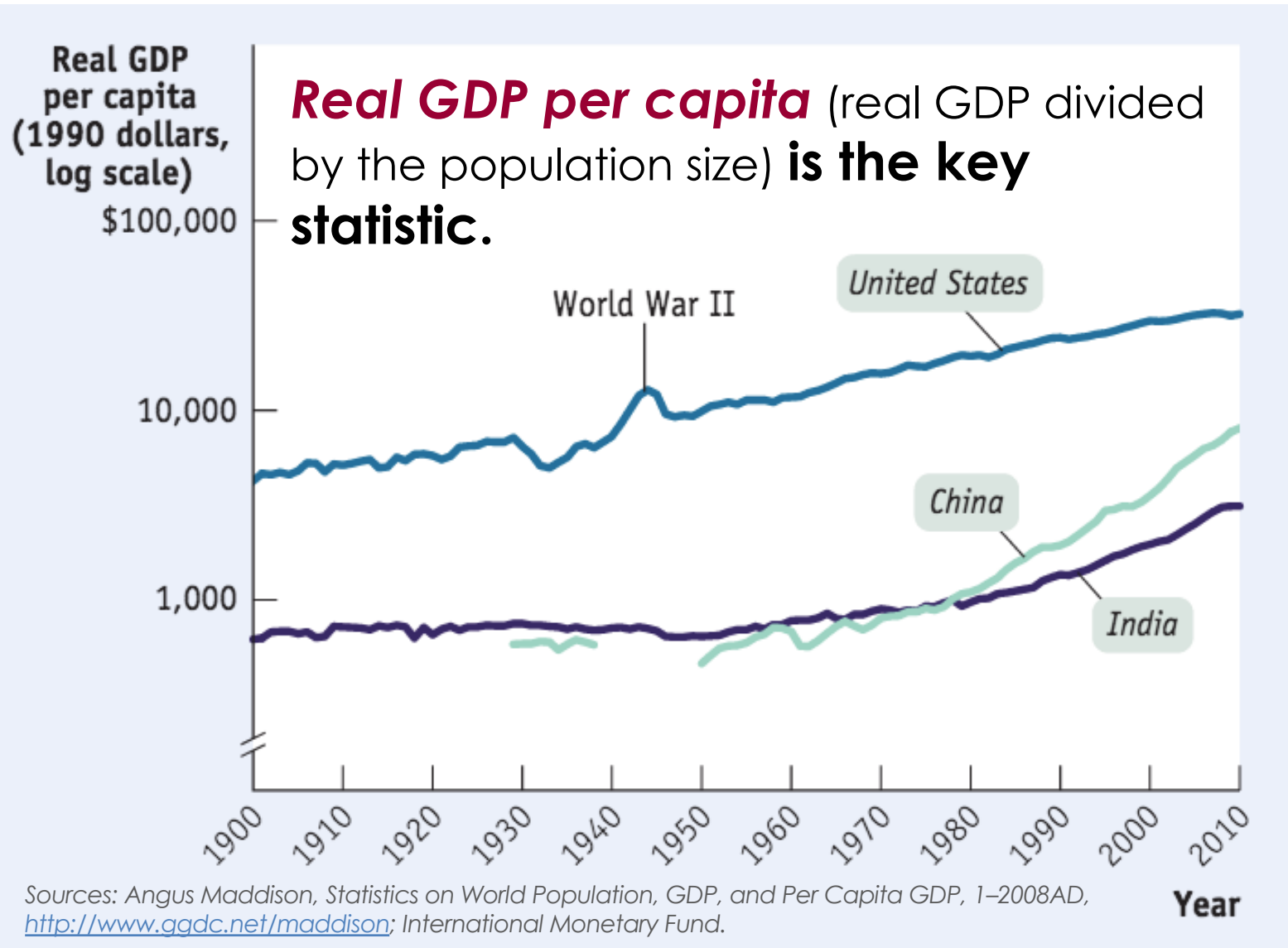
Cover of E J
Mishan's book
*The Costs of
Economic
Growth*, 1967



At TEDIndia, TED favorite Hans Rosling gives another lively and humorous talk [here](#) and graphs global economic growth since 1858. He also predicts the exact date that India and China will outstrip the United States. (15:50 minutes)



COMPARING ECONOMIES ACROSS TIME AND SPACE



LEARN BY DOING: PRACTICE QUESTION

Suppose real GDP for Macronesia is \$200 million in 2010. Furthermore, suppose population in Macronesia is 100,000 in 2010. If population increases to 105,000 in 2011 while GDP increases by 5%, then it must be true that real GDP per capita in Macronesia in 2011:

- a) increased.**
- b) decreased.**
- c) stayed constant.**
- d) may have increased, decreased, or remained constant.**

U.S. REAL GDP PER CAPITA

The U.S. economy produces more than seven times as much per person as in 1900.

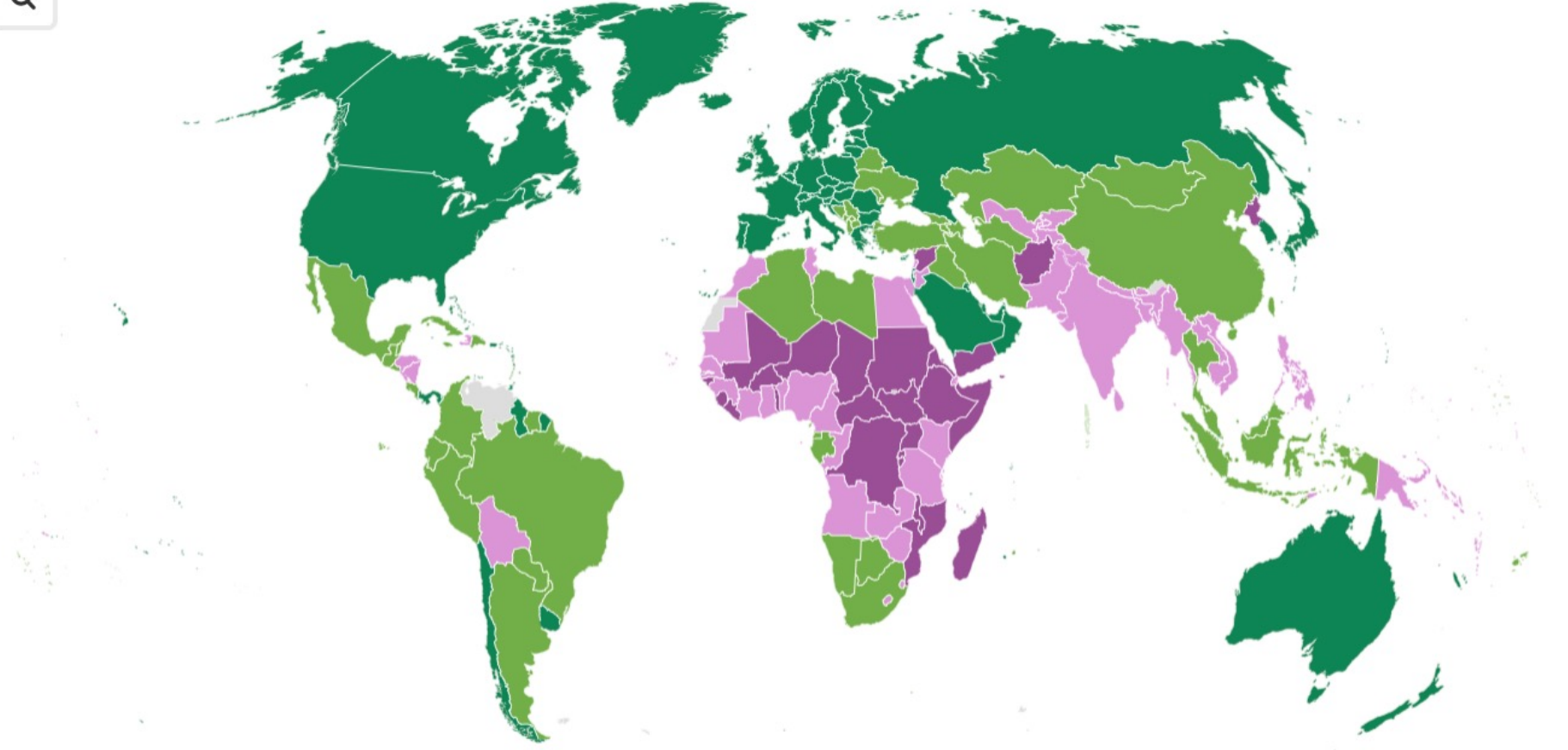
Year	Percentage of 1900 real GDP per capita	Percentage of 2010 real GDP per capita
1900	100%	13%
1920	136	18
1940	171	23
1980	454	60
2000	696	92
2010	758	100

Sources: Angus Maddison, *Statistics on World Population, GDP, and Per Capita GDP, 1–2008AD*, <http://www.ggdc.net/maddison>; Bureau of Economic Analysis.

INCOMES AROUND THE WORLD, 2023

**The United States has grown quickly, while some nations have stalled.
50% of the world's population lives in countries whose standard of living is lower than the United States' in 1900.**

High Income Upper-middle Income Lower-middle Income Low Income Not Classified



THE RULE OF 70: THE MAGIC OF COMPOUNDING

Even small differences in growth rates get magnified over time.

The *rule of 70*:

$$\text{Doubling time for a variable} = \frac{70}{\% \text{ growth rate}}$$

Example: If real GDP per capita is growing at an annual growth rate of 3.5%, it will double in:

$$\frac{70}{3.5} = 20 \text{ years.}$$

The moral? Small improvements in growth add up fast (the power of compounding).

LEARN BY DOING: PRACTICE QUESTION

Suppose that real GDP per capita grows at 2% per year. How many years will it take for real GDP per capita to approximately double? It will take:

- a) 70 years because of the rule of 70.**
- b) 140 years, because the rule of 70 states that a variable will double in 70 years if the variable has an annual growth rate of 1%; therefore, a variable growing at 2% will take twice as long to double.**
- c) 35 years, because $70/2 = 35$**
- d) 50 years, since at 2% per year it takes 50 years to reach 100% more than the initial real GDP per capita.**

LEARN BY DOING: PRACTICE QUESTION

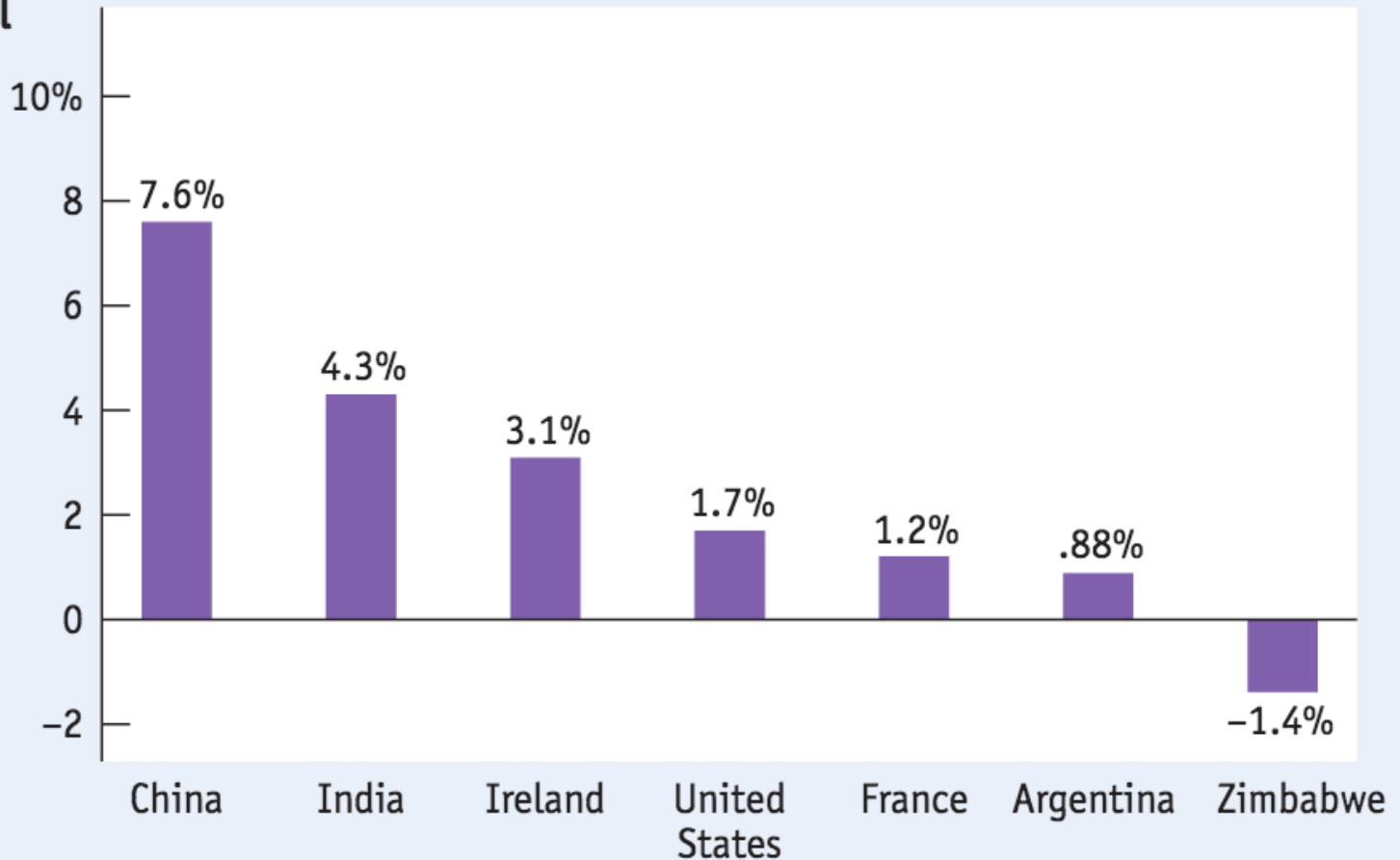
Let's figure out how long it will take for the average Indian to be as wealthy as the average Western European is today. *Note that all numbers are adjusted for inflation.* India's GDP per capita is \$3,000, and let's assume real output per person grows at 5% per year. Using the rule of 70, how many years will it take for India to reach Italy's current level of GDP per capita, about \$24,000 per year?

- a) 42 years
- b) 14 years
- c) 28 years
- d) 12 years

COMPARING RECENT GROWTH RATES

**China, India and Ireland saw impressive growth.
Zimbabwe, once the breadbasket of Africa, collapsed.**

Average annual
growth rate
of real GDP
per capita,
1980–2013





India's high rate of economic growth since 1980 has raised living standards and led to the emergence of a rapidly growing middle class. Will **lack of education and infrastructure** hold back future growth?

THE SOURCES OF LONG-RUN GROWTH

The crucial importance of productivity

Labor productivity (often referred to simply as productivity): output per worker



EXPLAINING GROWTH IN PRODUCTIVITY



Increase in physical capital



Physical capital: human-made resources, such as buildings and machines

EXPLAINING GROWTH IN PRODUCTIVITY



Increase in human capital



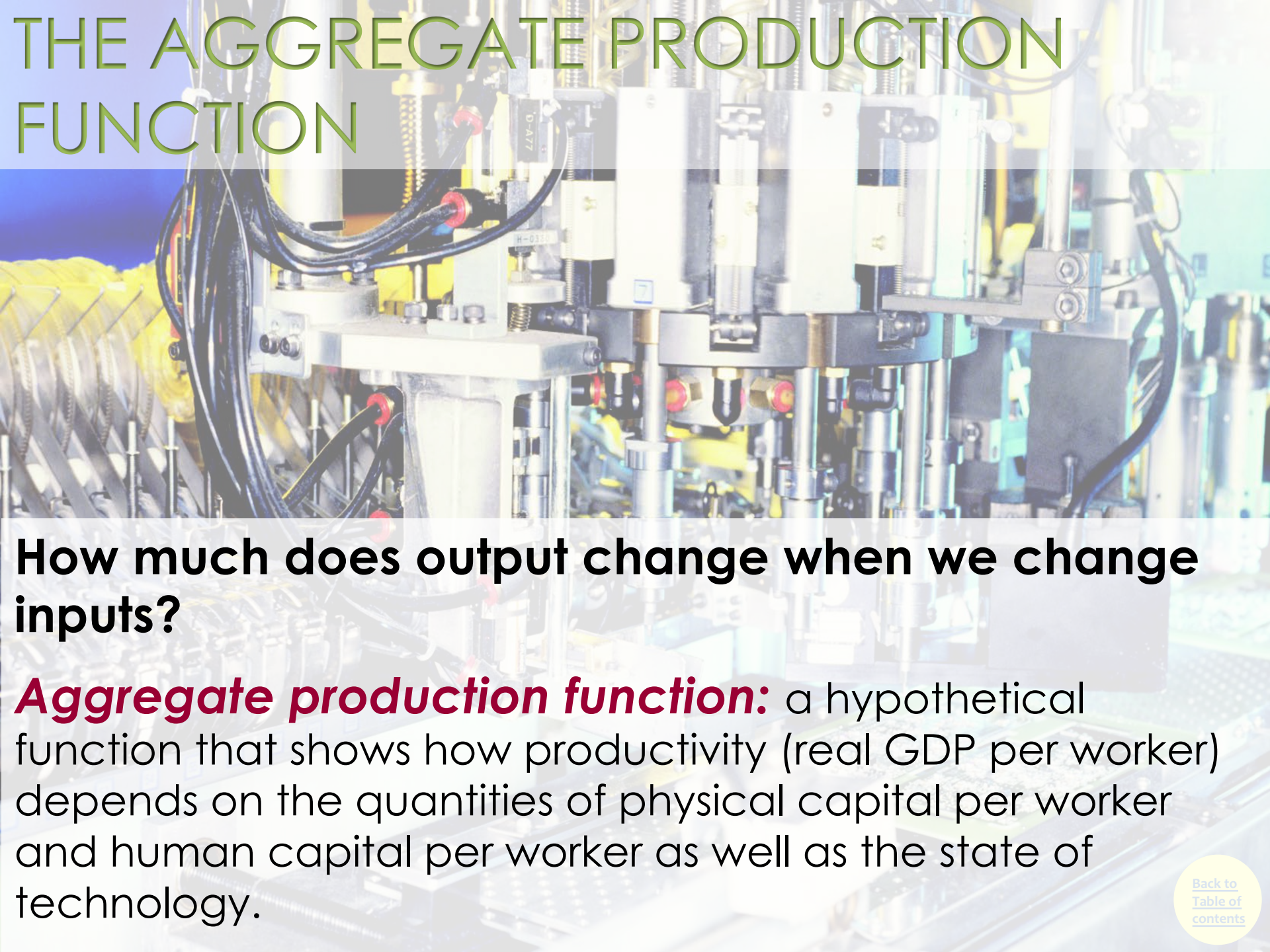
Human capital: the improvement in labor created by the education and knowledge embodied in the workforce.

EXPLAINING GROWTH IN PRODUCTIVITY

Technological progress: an advance in the technical means of production in goods & services.



THE AGGREGATE PRODUCTION FUNCTION



How much does output change when we change inputs?

Aggregate production function: a hypothetical function that shows how productivity (real GDP per worker) depends on the quantities of physical capital per worker and human capital per worker as well as the state of technology.

ACCOUNTING FOR GROWTH: THE AGGREGATE PRODUCTION FUNCTION

But can we be more specific? How much does output change when we change inputs?

For instance, China and India may have a production function like this (Brookings Institution estimates):

GDP per worker = $T \times (\text{physical capital per worker})^{0.4} \times (\text{human capital per worker})^{0.6}$

If China had more physical capital than India, that would explain its quicker growth.

THE AGGREGATE PRODUCTION FUNCTION

There are some well-known patterns.

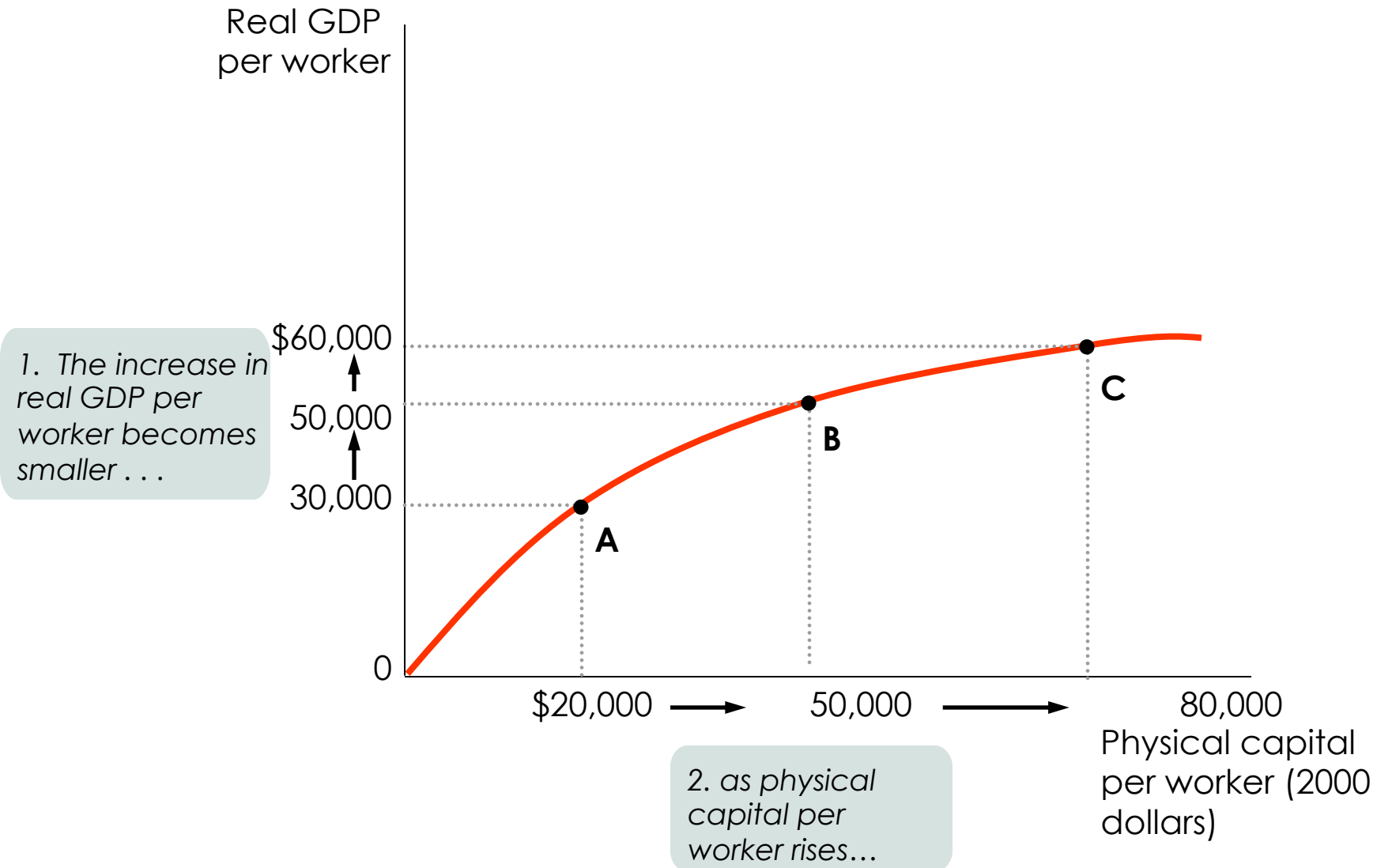
Diminishing returns to physical capital:

holding the amount of human capital per worker and the state of technology fixed, **each successive increase in the amount of physical capital per worker leads to a smaller increase in productivity.**



A second computer improves his productivity, but not by as much as the first computer did.

ACCOUNTING FOR GROWTH: THE AGGREGATE PRODUCTION FUNCTION



TECHNOLOGY AND PRODUCTIVITY

So additional amounts of physical capital are less productive *when the amount of human capital per worker and the technology are held fixed.*

But this assumes all other things equal.

Diminishing returns may disappear if we **increase the amount of human capital per worker** or **improve the technology** or **both** as the amount of physical capital per worker is increased.



TOTAL FACTOR PRODUCTIVITY

Real GDP per worker
(constant dollars)

\$120,000

60,000

50,000

30,000

0

\$20,000

50,000

80,000

Physical capital per
worker
(constant dollars)

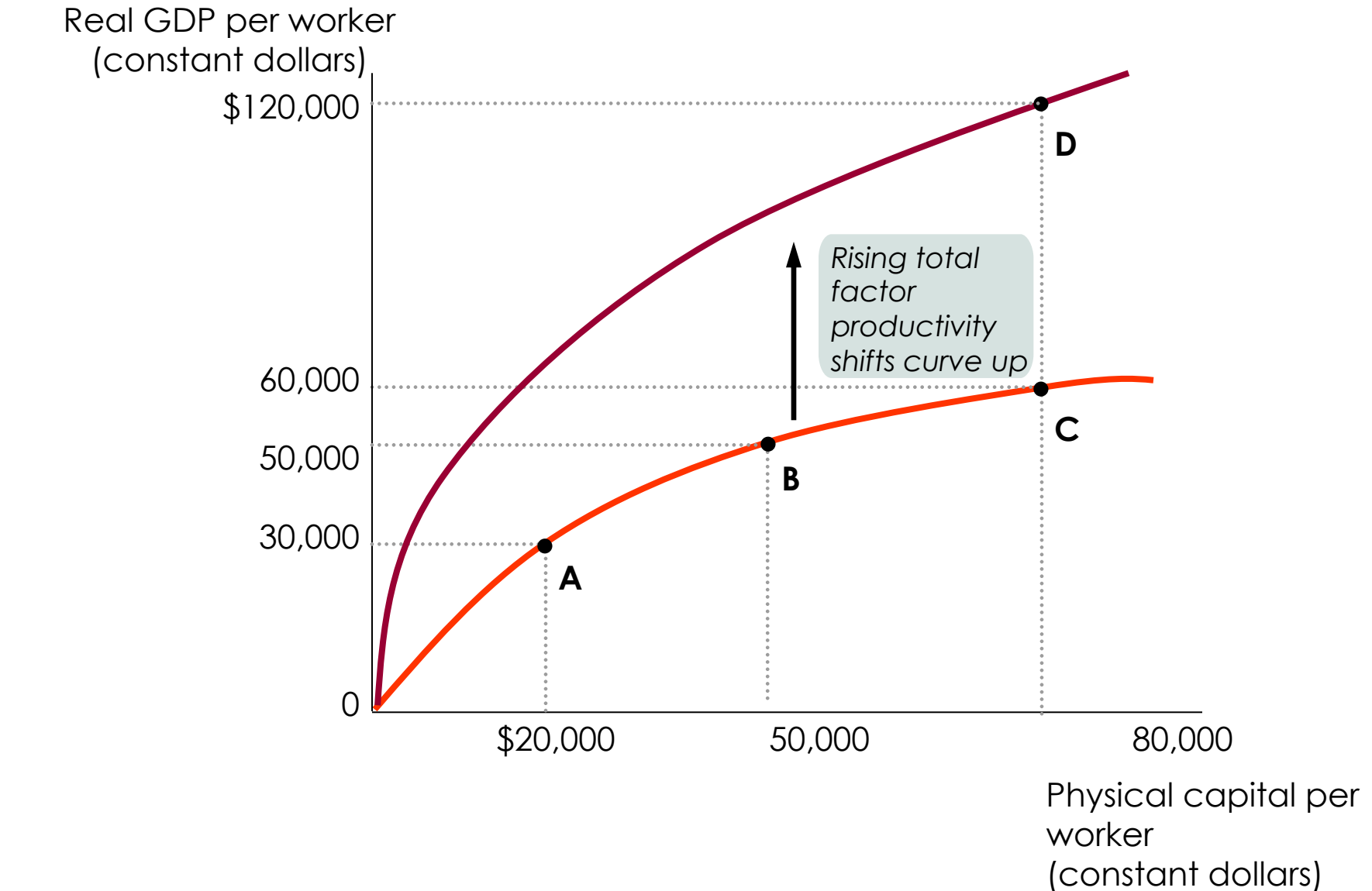
Rising total
factor
productivity
shifts curve up

A

B

D

C



GROWTH ACCOUNTING

How do we tell what caused the growth (and help develop future policy)?

Growth accounting: estimates of the contribution of each major factor in the aggregate production function to economic growth.

Total factor productivity: the amount of output that can be produced with a given amount of factor inputs.

(When total factor productivity increases, the economy can produce **more output** with the **same quantity of physical capital, human capital, and labor.**)

TECHNOLOGY AND PRODUCTIVITY

Increases in total factor productivity are crucial for growth.

And increases in total factor productivity likely measure the economic effects of technological progress.

So technological change is crucial to economic growth.

WHAT ABOUT NATURAL RESOURCES?



In the modern world, **human or physical capital matter more** than natural resources for the great majority of countries.

IS THE END OF ECONOMIC GROWTH IN SIGHT?

Economist **Robert Gordon** argues that new information technologies won't create as much growth as the five big innovations:

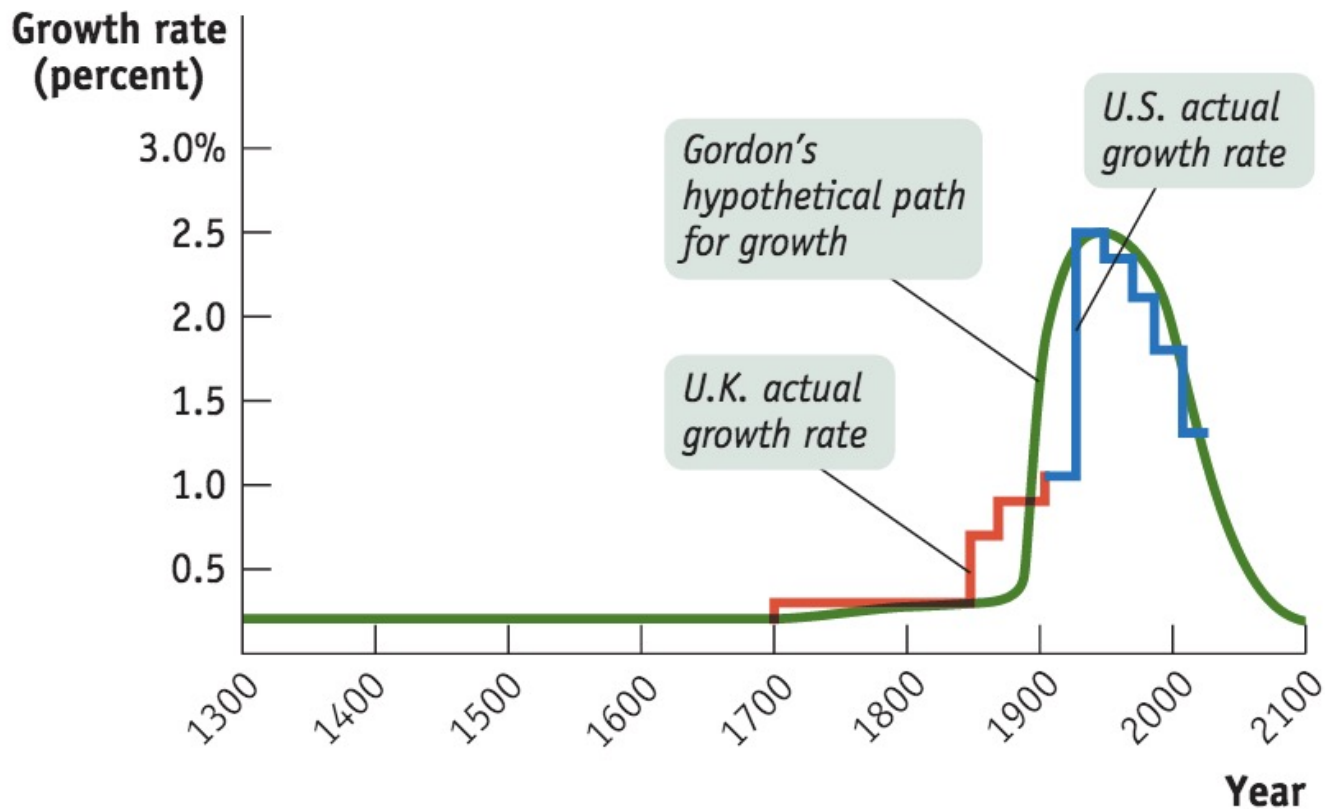
1. *Electricity*
2. *The internal combustion engine*
3. *Running water and central heating*
4. *Modern chemistry*
5. *Mass communication, movies, and telephones*

Is it too soon to tell?

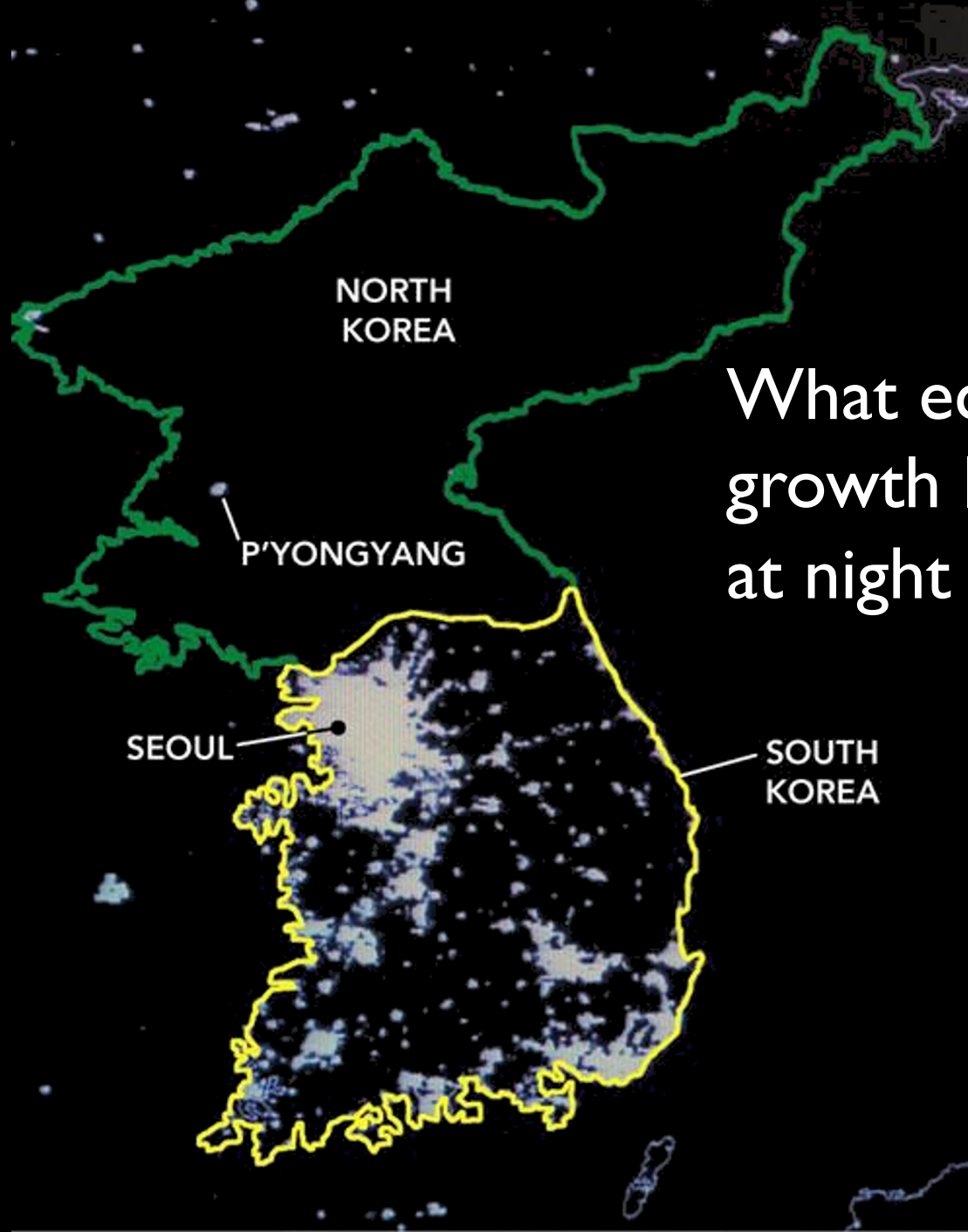
ECONOMICS IN ACTION

IS THE END OF ECONOMIC GROWTH IN SIGHT?

Robert Gordon's predictions vs. data



Source: Data from Robert J. Gordon.



What economic growth looks like at night

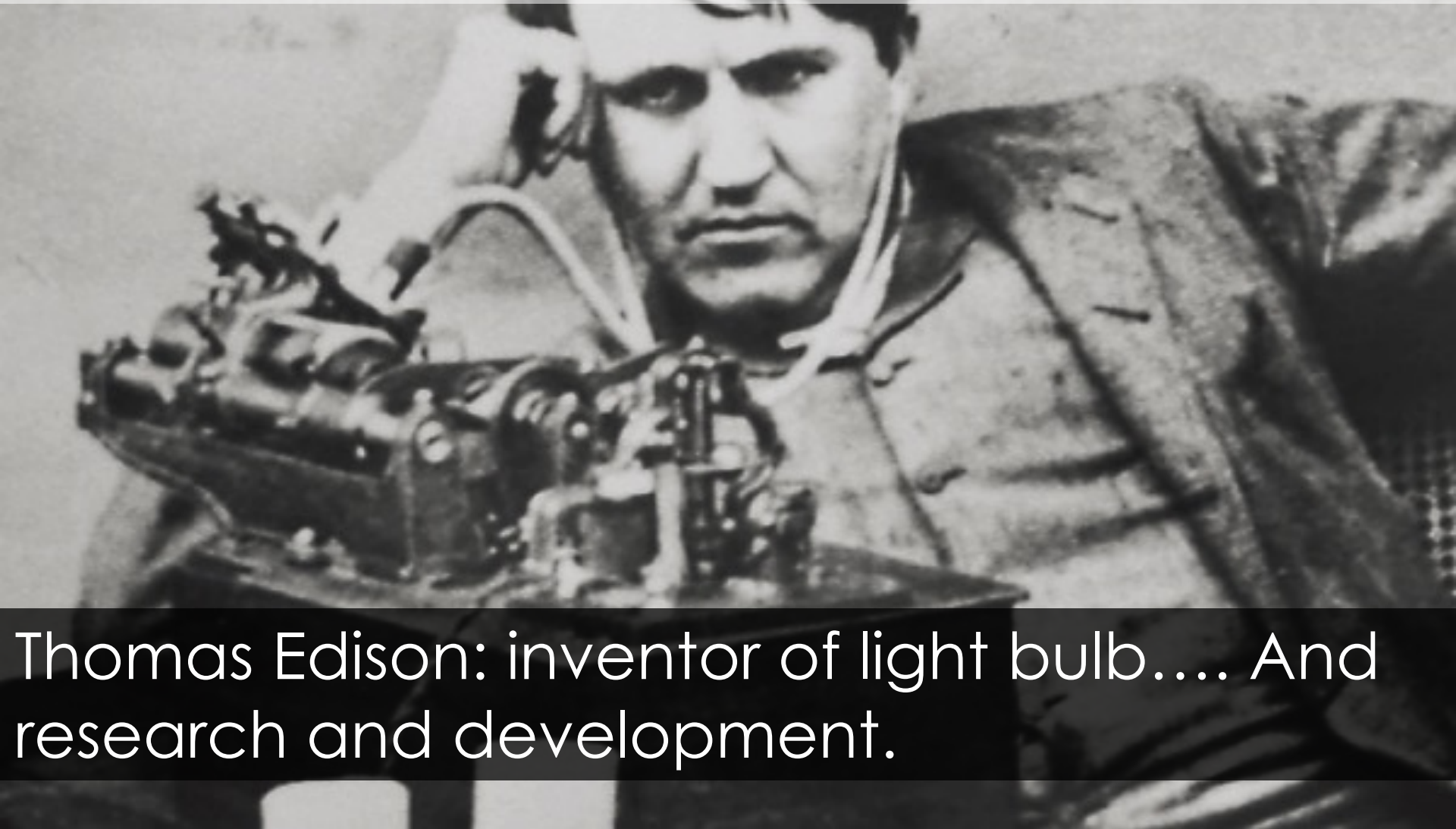
WHY GROWTH RATES DIFFER

Savings and investment spending
Education
Research and development



RESEARCH AND DEVELOPMENT

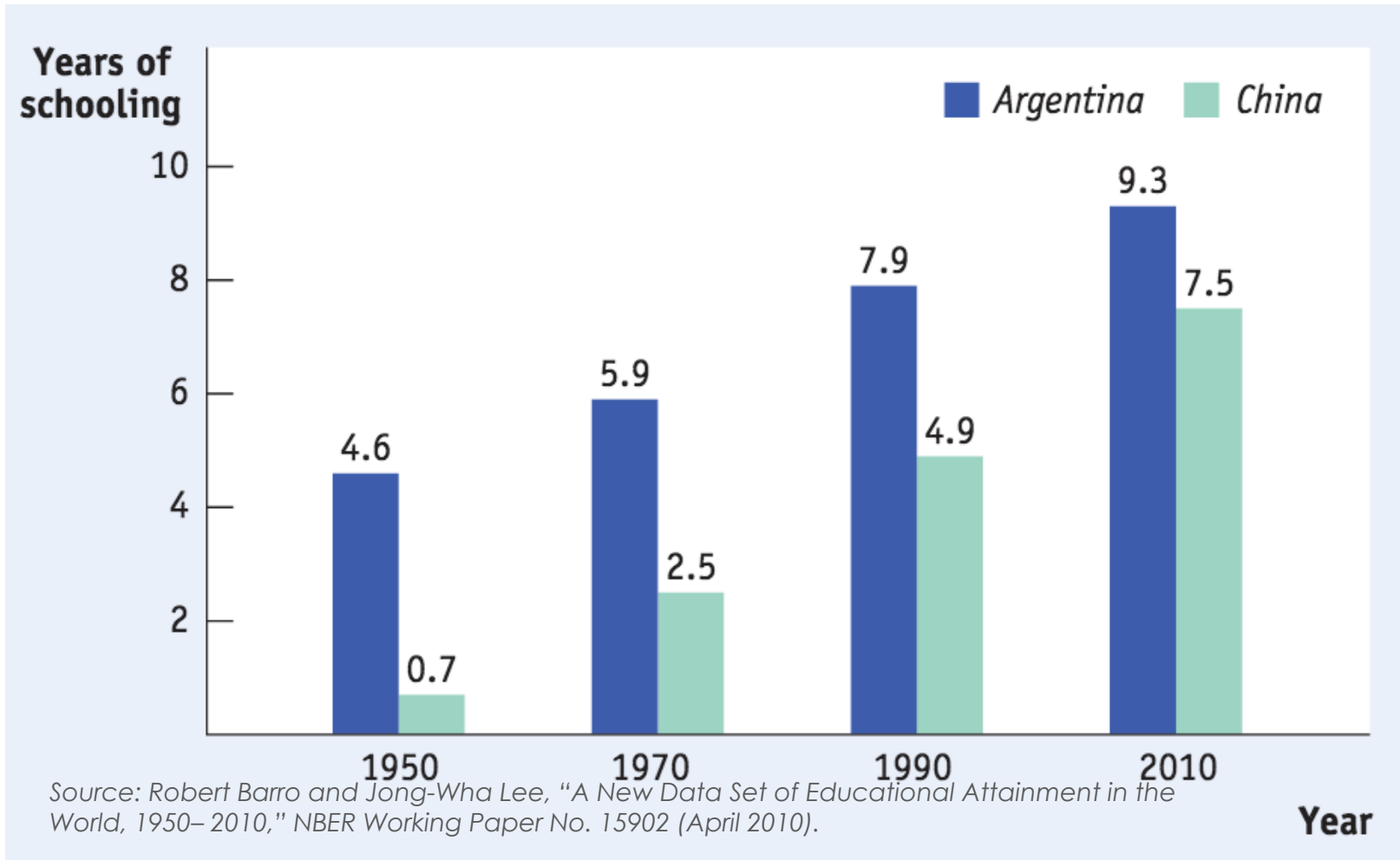
Research and development (R&D): spending to create and implement new technologies



Thomas Edison: inventor of light bulb.... And research and development.

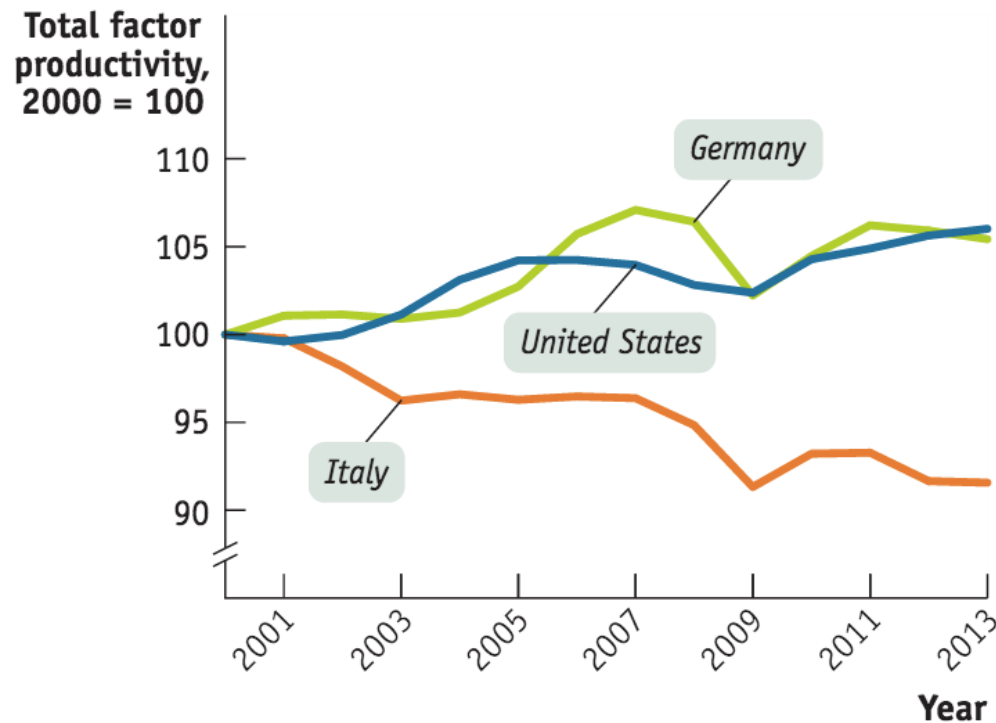
CHINA'S STUDENTS ARE CATCHING UP

China's success at adding human capital is one key to its spectacular long-run growth.



WHAT'S THE MATTER WITH ITALY?

Is Italy's productivity slump due to rigid labor markets, political weakness or slow information technology infrastructure?



Source: The Conference Board Total Economy Database, January 2014

THE ROLE OF GOVERNMENT IN PROMOTING ECONOMIC GROWTH

Government policies

1. GOVERNMENT SUBSIDIES TO INFRASTRUCTURE

Infrastructure: roads, power lines, ports, information networks, and other underpinnings for economic activity.

2. GOVERNMENT SUBSIDIES TO EDUCATION

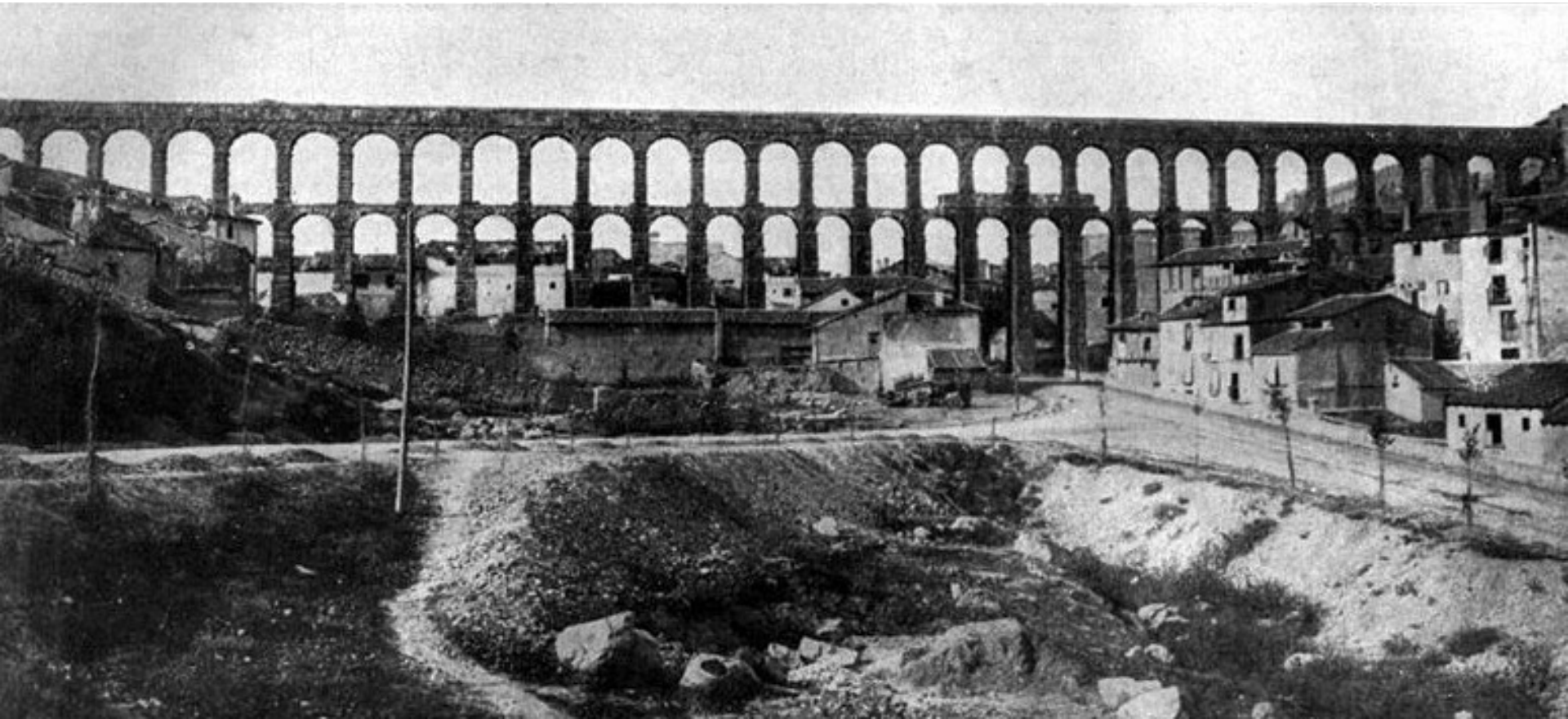
3. GOVERNMENT SUBSIDIES TO R&D

4. MAINTAINING A WELL-FUNCTIONING FINANCIAL SYSTEM

Protection of property rights

Political stability and good governance

INFRASTRUCTURE



**Roman aqueducts: ushering in an era of economic growth.
Aqueducts were sophisticated feats of engineering that enabled
population and industry to thrive.**

Roman Aqueduct in Segovia, Spain, built in second century A.D./C.E.

ECONOMICS *IN ACTION*

WHY DID BRITAIN FALL BEHIND?

There are many theories- but it's clear Britain fell behind the United States and Germany largely due to barriers to education.

WHAT CAUSED EAST ASIA'S MIRACLE?



Very high savings rates that allow business to borrow and add more physical capital per worker.

Very good basic education has permitted a rapid improvement in human capital.

Substantial technological progress.

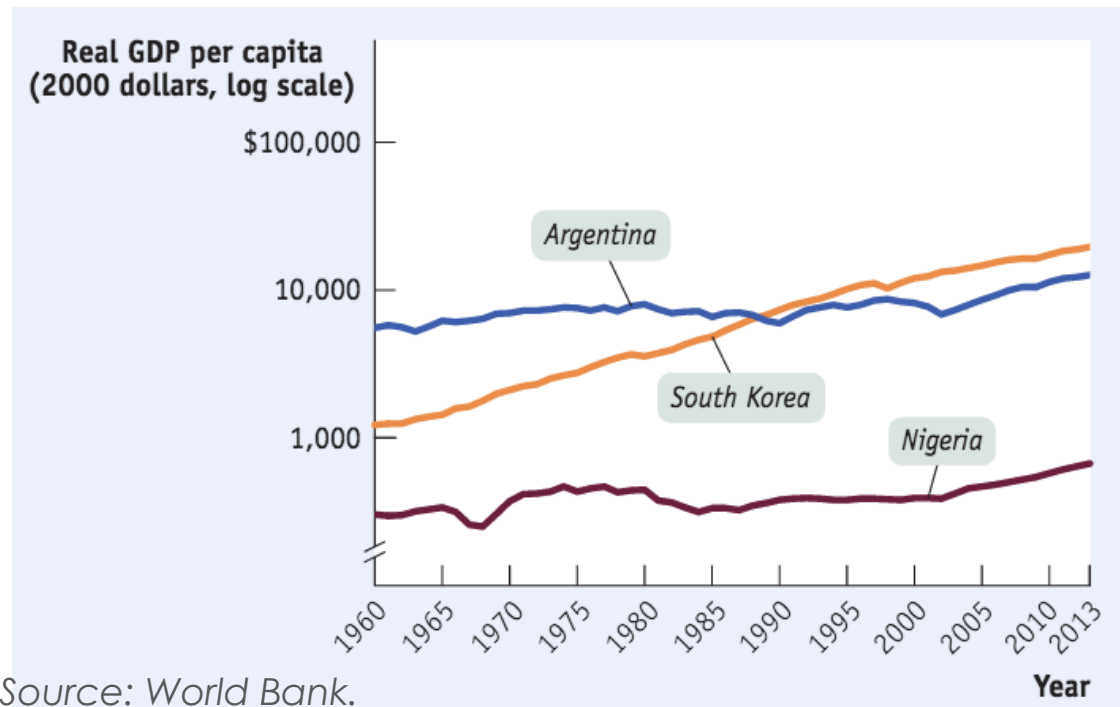
SUCCESS, DISAPPOINTMENT AND FAILURE

What's holding Argentina (and other Latin American nations) back?

Irresponsible government action that eroded savings through high inflation

Lack of emphasis on education

Political instability



AFRICA'S TROUBLES AND PROMISE

What's holding much of Africa back?

Government corruption

Civil wars & political instability

Unfavorable geography

A big question: Is Africa poor because its politics are unstable? Or vice versa?

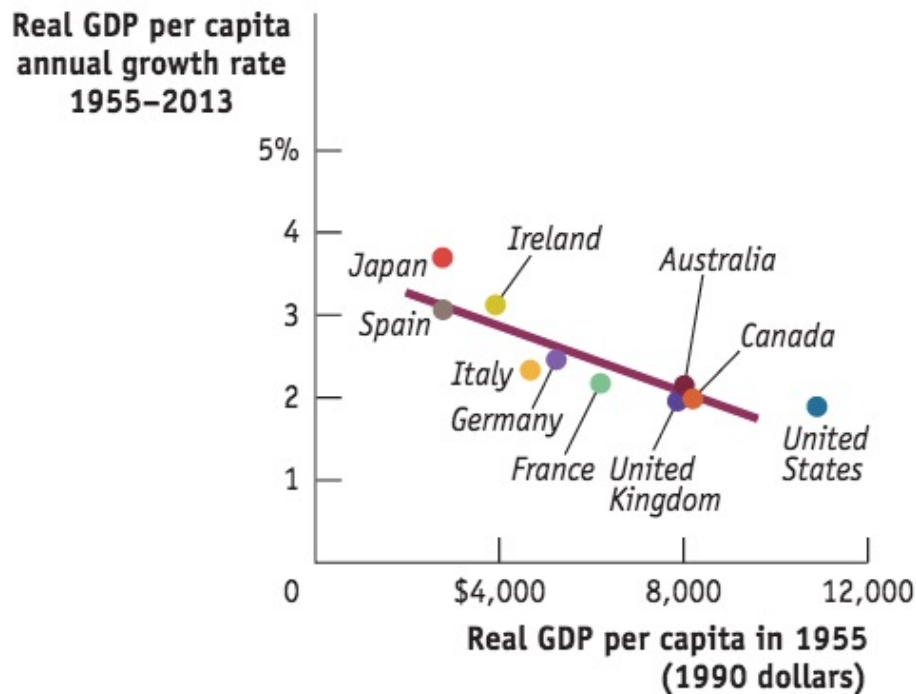
Good news: Growth rates are up in sub-Saharan African nations since 2011.



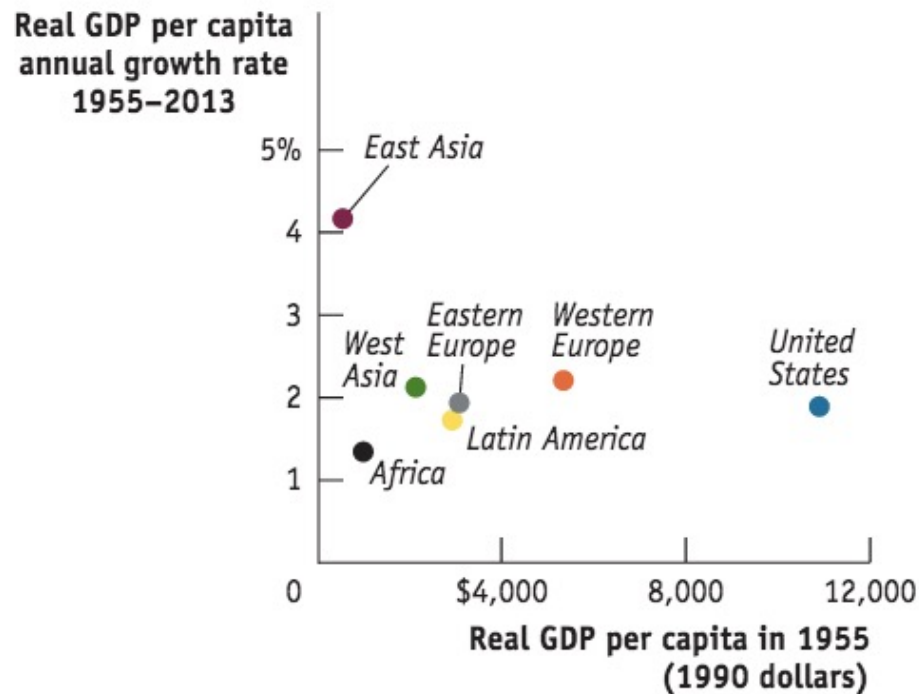
CONVERGENCE HYPOTHESIS

Convergence hypothesis: international differences in real GDP per capita tend to narrow over time.

(a) Convergence among Wealthy Countries...



(b) ... But Not for the World as a Whole



Sources: Angus Maddison, *Statistics on World Population, GDP, and Per Capita GDP, 1–2008AD*, <http://www.ggd.net/maddison>;
The Conference Board Total Economy Database™, January 2014, <http://www.conference-board.org/data/economydatabase>.

LEARN BY DOING: PRACTICE QUESTION

Economic growth can be especially fast:

- a) for countries playing catch-up with countries that already have high real GDP per capita.**
- b) for relatively poor countries if the convergence hypothesis holds true.**
- c) if the country is able to benefit from adopting the technological advances already used in advanced countries.**
- d) Answers (a), (b), and (c) are all true.**

LEARN BY DOING: APPLICATION VIDEO

At TEDIndia, TED favorite Hans Rosling gives another lively and humorous talk [here](#) and graphs global economic growth since 1858. He also predicts the exact date that India and China will outstrip the United States. (15:50 minutes)



IS WORLD GROWTH SUSTAINABLE?

Sustainable long-run economic growth: long-run growth that can continue in the face of the limited supply of natural resources and the impact of growth on the environment.

What is the impact of limited natural resources on long-run economic growth? Three important questions:

How large are the supplies of key natural resources?

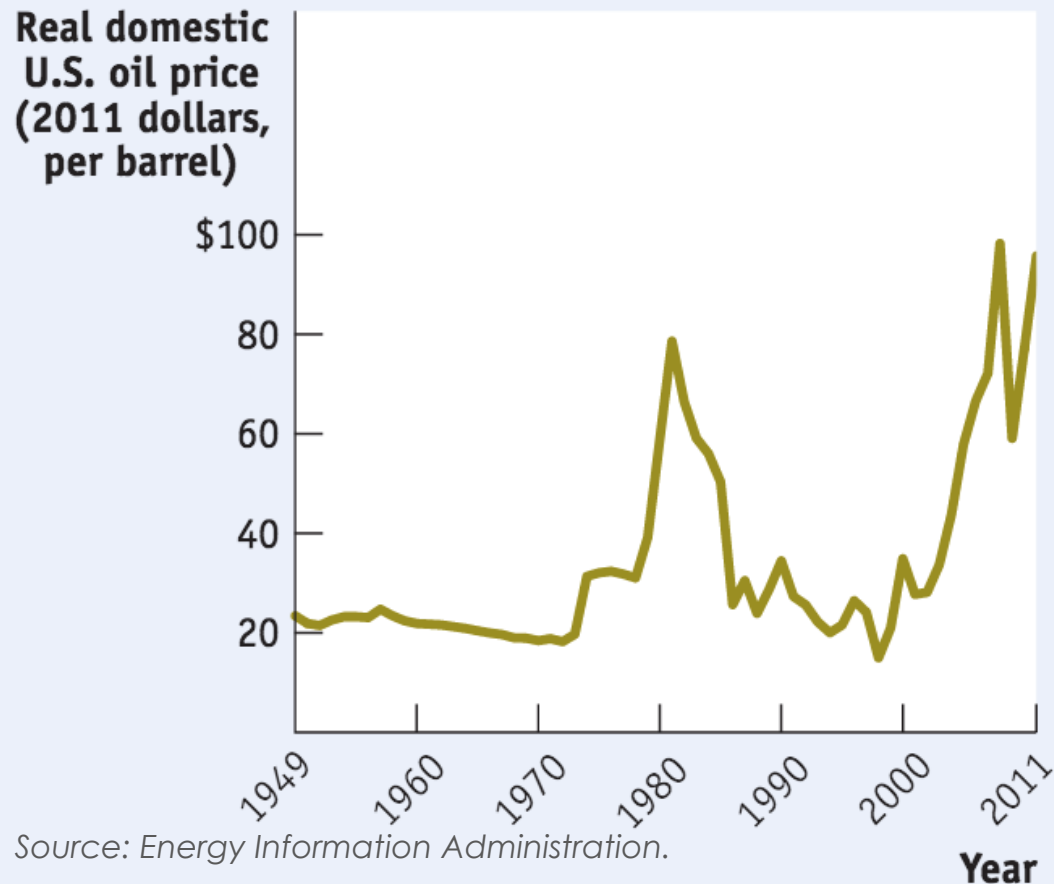
How effective will technology be at finding alternatives to natural resources?

Can long-run economic growth continue in the face of resource scarcity?

NATURAL RESOURCES AND GROWTH, REVISITED

How large are the supplies of key natural resources?

Geologists are figuring it out.



NATURAL RESOURCES AND GROWTH, REVISITED



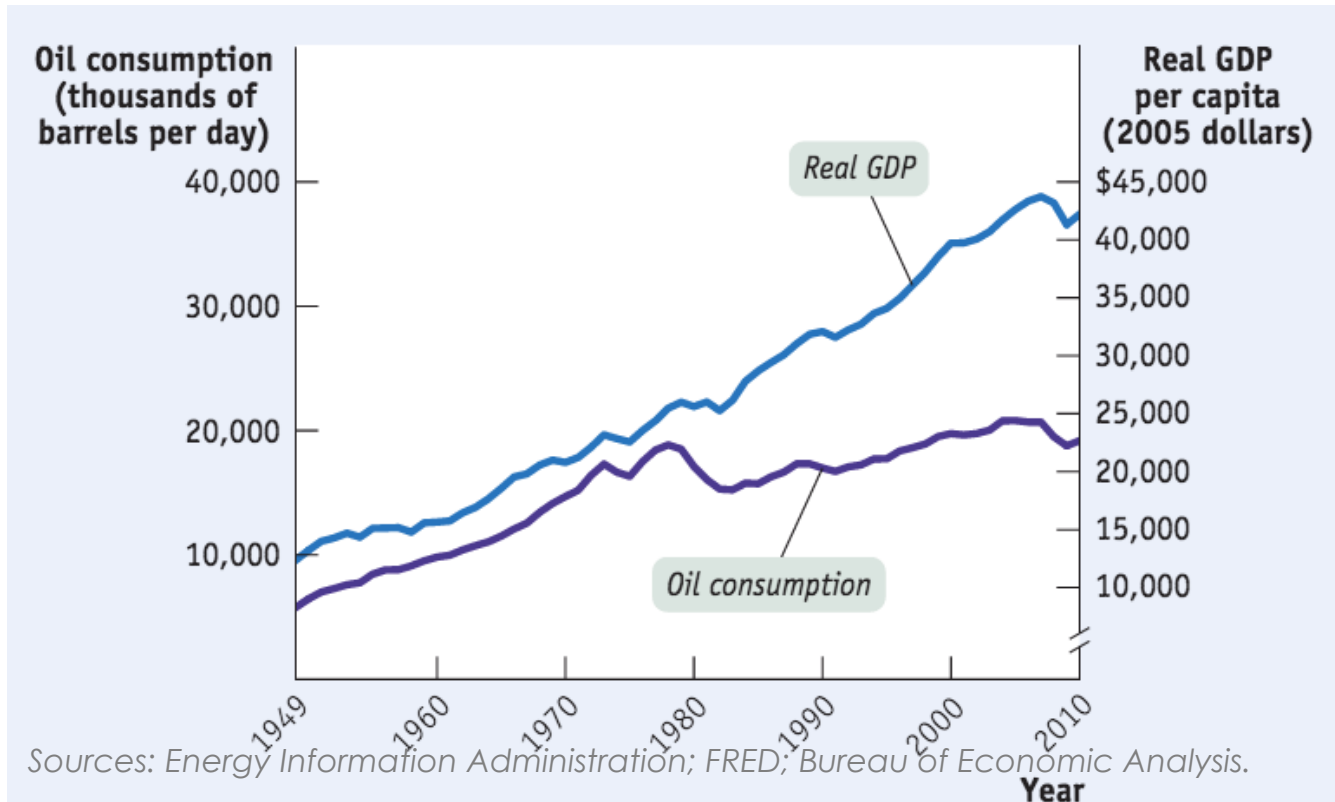
How effective will technology be at finding alternatives to natural resources?

Engineers are figuring it out.

NATURAL RESOURCES AND GROWTH, REVISITED

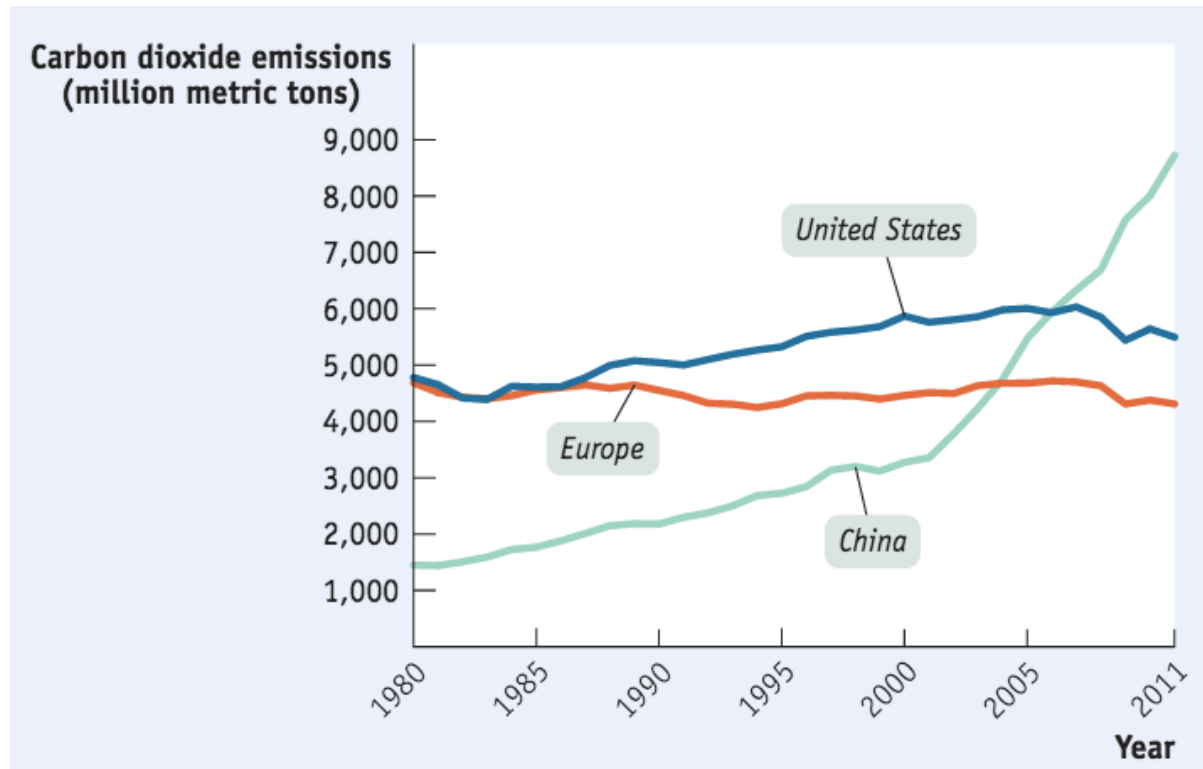
Can long-run economic growth continue in the face of resource scarcity?

A question for economists. Resource scarcity and higher prices don't necessarily cost growth.



ECONOMIC GROWTH AND THE ENVIRONMENT

Economic growth tends to increase the human impact on the environment.



Sources: Energy Information Administration; Bureau of Economic Analysis.

Unlike resource scarcity, environmental problems don't automatically provide incentives for changed behavior.

ECONOMIC GROWTH AND THE ENVIRONMENT

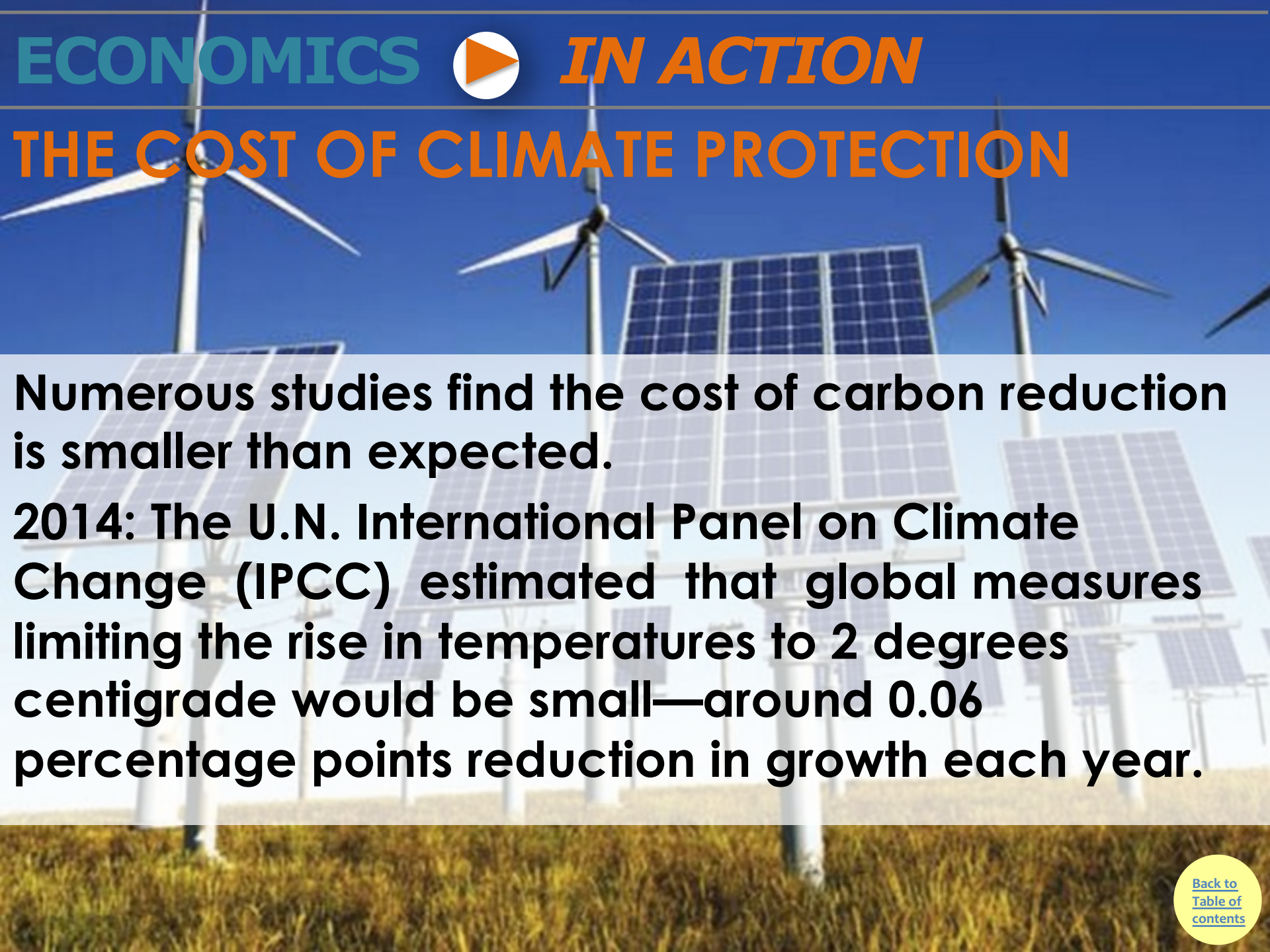
There is consensus in economics that government action is needed to deal with climate change.

Economists prefer market-based incentives, in the form of:

How much action is appropriate? We are uncertain about how fast climate change is happening and what impact the solutions will have.

Carbon tax (a tax per unit of carbon emitted) or

Cap and trade system in which the total amount of emissions is capped and producers must buy licenses to emit greenhouse gases.



ECONOMICS ***IN ACTION***

THE COST OF CLIMATE PROTECTION

Numerous studies find the cost of carbon reduction is smaller than expected.

2014: The U.N. International Panel on Climate Change (IPCC) estimated that global measures limiting the rise in temperatures to 2 degrees centigrade would be small—around 0.06 percentage points reduction in growth each year.